

B. Sc. 5th Semester (Honours) Examination, 2025
Computer Science

Course ID: 51516

Course Code: SH/CSC/503/DSE-1

Numerical Methods - (Theory)
(NEW CBCS Syllabus 2022-23)

Time: 1 Hour 15 Mins

Full Marks: 25

The figures in the margin indicate full marks

UNIT I

1. Answer any 5 (five) of the following questions:

(5 × 1 = 5)

- a) What is computer arithmetic?
- b) Express floating-point numbers in general form.
- c) What do you mean by iteration?
- d) Name two methods used for numerical integration.
- e) Distinguish between the first and second order derivatives.
- f) What is extrapolation?
- g) Why does Newton-Raphson method runs faster than bisection method?
- h) Distinguish between algebraic and transcendental equation.

[Turn Over]

UNIT II

2. Answer any 2 (two) of the following questions:

(2 × 5 = 10)

- a) Write a short note on different types of errors occurred in numerical analysis. 5
- b) Explain the method of false position for computing the real root of the equation $f(x) = 0$.
State an advantage of Secant method over Newton-Raphson method. 4+1=5
- c) The following values of x & y are given:

X: 1 2 3 4

Y: 3 11 31 69

Evaluate y (1.8).

- d) Using the fixed point iteration method determine a real root of $x - e^{-x} = 0$

UNIT III

3. Answer any 1 (one) of the following questions:

(1 × 10 = 10)

- a) Solve the following system of equations by Gauss Elimination Method:
 $7x + 2y + 4z = 16$

$$2x + y + 7z = 10$$

$$2x + 3y + 5z = 2$$

Discuss Newton's backward interpolation with a suitable example. 5+5=10

b) Solve the following by Euler's modified method

$$dy/dx = \log(x+y), y(0) = 2 \text{ at } x = 1.2 \text{ with } h = 0.2$$

Discuss the Lagrange's interpolation method. 5+5=10

B. Sc. 5th Semester (Honours) Examination, 2025 **2.**
Computer Science

Course ID: 51516

Course Code: SH/CSC/503/DSE-

Operations Research -(Theory)
(NEW CBCS Syllabus 2022-23)

Time: 1 Hour 15 Mins

Full Marks: 20

The figures in the margin indicate full marks

UNIT I

1. Answer any 5 (five) of the following questions:

(5 × 1 = 5)

- a) Name some creators of LPP.
- b) Why do we need null vector?
- c) Why do we need matrices?
- d) What do you mean by a basic feasible solution of a LPP?
- e) What do you mean by interior and boundary point of a set?
- f) What do you mean by a close set?
- g) Show that $\{x: |x| \leq 2\}$ is a convex set.
- h) What do you mean by a non-degenerate basic solution?

[Turn Over]

UNIT II

2. Answer any 2 (two) of the following questions:

(2 × 5 = 10)

a) Make a graphical representation of the following LP:

$$\text{Max } z = 2x_1 + x_2$$

$$\text{Subject to: } 4x_1 + 3x_2 \leq 12, 4x_1 + x_2 \leq 8, 4x_1 - x_2 \leq 8$$

$$x_1, x_2 \geq 0$$

b) Geometrically draw the feasible space, if any, given by the following LP and find out the extreme points of the feasible region:

$$\text{Min } z = 4x_1 + 5x_2$$

$$\text{Subject to: } x_1 + 2x_2 \geq 2$$

$$5x_1 + 3x_2 \leq 15$$

c) What do you mean by linearly independent and linearly dependent variables? Show that following vectors form a basis for E' : (3, 0, 2), (7, 0, 9) and (4, 1, 2)

$$2+3=5$$

d) Transform the following LP in standard maximization form:

$$\text{Min } z = 3x_1 - 4x_2 - x_3$$

$$\text{Subject to: } x_1 - 4x_2 - 5x_3 \geq 5, x_1 + 3x_2 - 4x_3 < 12, 2x_1 + x_2 + x_3 \leq 20$$

Where $x_1 \geq 0$, x_2 and x_3 are unrestricted in sign.

UNIT III

3. Answer any 1 (one) of the following questions:

(1 × 10 = 10)

a) An agricultural farm has 180 tons of Nitrogen fertilizers, 50 tons of Phosphate and 220 tons of Potash. It will be able to sell 3:3:4 mixtures of these substances at a profit of Rs 15 per ton and 2:4:2 mixtures at a profit of Rs 12 per ton respectively. Form a LPP to show how many tones of the two mixtures should be prepared to obtain the maximum profit.

Solve the resulting LPP by using any method. **5+5=10**

b) Solve the following LPP by simplex method:

$$\text{Max } z = 4x_1 + 10x_2$$

$$\text{Subject to: } 2x_1 + x_2 \leq 50, 2x_1 + 5x_2 \leq 100, 2x_1 + 3x_2 \leq 30, \\ x_1, x_2 \geq 0$$

B. Sc. 5th Semester (Honours) Examination, 2025

Computer Science

Course ID: 51516

Course Code: SH/CSC/503/DSE-1

Numerical Methods - (Theory)

(OLD CBCS Syllabus 2017-18)

Time: 1 Hour 15 Mins

Full Marks: 25

The figures in the margin indicate full marks

UNIT I

Answer any 5 (five) of the following questions:

(5 × 1 = 5)

- a) What is computer arithmetic?
- b) What are the different types of errors occur while doing numerical computations?
- c) Find absolute error if the number $X=0.00599826$ is truncated to four decimal digits.
- d) Name two methods used for numerical integration.
- e) Distinguish between round off errors & truncation errors.
- f) What is extrapolation?
- g) Why does Newton-Raphson method run faster than bisection method?
- h) Distinguish between algebraic and transcendental equation.

[Turn Over]

UNIT II

2. Answer any 2 (two) of the following questions:

3
(2 × 5 = 10)

- a) Write a short note on different types of errors occurred in numerical analysis.
- b) Explain the method of false position for computing the real root of the equation $f(x) = 0$.
State an advantage of Secant method over Newton-Raphson method. 4+1:
- c) The following values of x & y are given:

X: 1 2 3 4

Y: 3 11 31 69

Evaluate y (1.8). b

- d) Discuss any one method for numerical integration

UNIT III

3. Answer any 1 (one) of the following questions:

(1 × 10 = 10)

- a) Solve the following system of equations by Gauss Elimination Method:

$$7x + 2y + 4z = 16$$

$$2x + y + 7z = 10$$

$$2x + 3y + 5z = 2$$

Discuss Newton's forward interpolation with a suitable example. 5+5=10

- b) Solve the following by Euler's modified method

$$dy/dx = x^2 + y^2, y(1) = 0 \text{ at } x=2 \text{ with } h=0.2$$

1 Discuss the Lagrange's interpolation method. 5+5=10

B. Sc. 5th Semester (Honours) Examination, 2025
Computer Science

Course ID: 51516

Course Code: SH/CSC/503/DSE

Operations Research -(Theory)
(OLD CBCS Syllabus 2017-18)

Time: 1 Hour 15 Mins

Full Marks:

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UNIT I

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- f) What do you mean by a closed set?
- g) Show that $\{x: |x| \leq 2\}$ is a convex set.
- h) What do you mean by a non-degenerate basic solution?

UNIT II

2. Answer any 2 (two) of the following questions:

(2 × 5 = 10)

- a) Make a graphical representation of the following LP:
Max $z = 2x_1 + x_2$
Subject to: $4x_1 + 3x_2 \leq 12$, $4x_1 + x_2 \leq 8$, $4x_1 - x_2 \leq 8$
 $x_1, x_2 \geq 0$
- b) Geometrically draw the feasible space, if any, given by the following LP and find out the extreme points of the feasible region:
Min $z = 4x_1 + 5x_2$
Subject to: $x_1 + 2x_2 \geq 2$
 $5x_1 + 3x_2 \leq 15$
- c) What do you mean by linearly independent and linearly dependent variables? Show that following vectors form a basis for E' : (3, 0, 2), (7, 0, 9) and (4, 1, 2) 2+3=5
- d) Transform the following LP in standard maximization form:
- Min $z = 3x_1 - 4x_2 - x_3$
Subject to: $x_1 - 4x_2 - 5x_3 \geq 5$, $x_1 + 3x_2 - 4x_3 < 12$, $2x_1 - x_2 + x_3 \leq 20$
Where $x_1 \geq 0$, x_2 and x_3 are unrestricted in sign.

UNIT III

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(1 × 10 = 10)

a) An agricultural farm has 180 tons of Nitrogen fertilizer, 50 tons of Phosphate and 220 tons of Potash. It will be able to sell 3:3:4 mixtures of these substances at a profit of Rs 15 per ton and 2:4:2 mixtures at a profit of Rs 12 per ton respectively. Form a LPP to show how many tones of the two mixtures should be prepared to obtain the maximum profit.

Solve the resulting LPP by using any method. 5+5=10

b) Solve the following LPP by simplex method:

$$\text{Max } z = 4x_1 + 10x_2$$

$$\text{Subject to: } 2x_1 + x_2 \leq 50, 2x_1 + 5x_2 \leq 100, 2x_1 + 3x_2 \leq 30,$$

$$x_1, x_2 \geq 0$$

Bankura Sammilani College

Dept. Of Computer Science

Internal Assessment Examination - 2024

Semester – V (Numerical Analysis)

Time 30 Minutes

F.M – 10

Answer any 5 questions-

5x2=10

1. Define truncation error.
2. Write advantage and disadvantage of Regula-falsi method over other methods of solving algebraic and transcendental equations.
3. Evaluate $\int_1^2 (2x + 3)dx$ using Simpson's 1/3 rule. (Take $h = 0.2$)
4. Evaluate $\int_1^2 (x^2 + 3)dx$ using Trapezoidal rule. ($n = 5$)
5. When NR method fails and when it diverges.
6. Solve the following system of linear equations by Gauss elimination method.
$$\begin{aligned} 2x + 3y + z &= 9 \\ x + 2y + 3z &= 6 \\ 3x + y + 2z &= 8 \end{aligned}$$
7. Round-off the following numbers to 4 significant figures.
 - a. 0.0056812
 - b. 40.3585

Of

B. Sc. 5th Semester (Honours) Examination, 2025
Computer Science

Course ID: 51526

Course Code: SH/CSC/503/DSE-1

Operation Research (Practical)
(NEW CBCS Syllabus 2022-2023)

Time: 2 Hrs

Full Marks: 15

The figures in the margin indicate full marks

(Experiment = 10, LNB + Viva Voce = 05)

1. Perform any one experiment: (1 × 10 = 10)

1. Solve the following LPP using simplex method-
Min. $Z = x - 3y + 2z$ subject to $3x - y + 3z \leq 7$,
 $-2x + 4y \leq 12$, $-4x + 3y + 8z \leq 10$
2. Solve the following LPP using two phase method -
Max. $Z = 3x - y$ subject to $2x + y \geq 2$, $x + 3y \leq 2$, $y \leq 4$, and $x, y \geq 0$.
3. Use Big-M method to solve the following LP problem -
Max. $Z = -2x - y$ subject to $3x + y = 3$, $4x + 3y \geq 6$, $x + 2y \leq 4$ and $x, y \geq 0$.
4. Solve the following problem by Dual-Simplex method -
Min. $Z = 2x + y$ subject to $3x + y \geq 3$, $4x + 3y \geq 6$, $x + 2y \geq 3$ and $x, y \geq 0$.

[Turn Over]

5. Solve the following LPP by revised simplex method

$$\text{Max } z. = x + 2y \text{ subject to } x + 2y \leq 3, x + 2y \leq 1 \text{ and } x, y \geq 0$$

B. Sc. 5th Semester (Honours) Examination, 2025
Computer Science

Course ID: 51526

Course Code: SH/CSC/503/DSE-1

Numerical Methods (Practical)
(NEW CBCS Syllabus 2022-2023)

Time: 2 Hrs

Full Marks: 15

The figures in the margin indicate full marks

(Experiment = 10, LNB + Viva Voce= 05)

1. Perform any one experiment: (1 × 10 = 10)

1. Find a real root of $x^3 + 5x - 5 = 0$ to two decimal places using bisection method.
2. Evaluate $\int_1^2 x^2 dx$ using simpson's 1/3 rule taking $h=0.2$.
3. Evaluate $\int_1^2 x^2 dx$ using trapezoidal rule taking $h=0.1$.
4. Find a real root of $x^3 - x - 5 = 0$ to 3 decimal places using method of false position
5. Find a real root of $x^3 - x - 5 = 0$ to 3 decimal places using N-R method.
6. Solve the Initial Value Problem:
 $\frac{dy}{dx} = 6 - 2\left(\frac{y}{x}\right), y(3) = 1$ using any R-K method of your choice.

[Turn Over]

7. Given the following data table, construct the Lagrange interpolation polynomial $f(x)$, to fit the data and find $f(1.25)$

i	0	1	2	3
x_i	0	1	2	3
$f(x_i)$	1	2.25	3.75	4.25

8. Using Gaussian elimination method solve the following system of linear equations:

$$X+Y+Z=6$$

$$X+2Y-3Z=-4$$

$$4X+5Y+Z=17$$

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Computer Science

Course ID: 51526

Course Code: SH/CSC/503/DSE-1

Numerical Methods (Practical)
(OLD CBCS Syllabus 2017-2018)

Time: 2 Hrs

Full Marks: 15

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(Experiment = 10, LNB + Viva Voce = 05)

1. Perform any one experiment: (1 × 10 = 10)

1. Find a real root of $x^3 + 5x - 5 = 0$ to two decimal places using bisection method.
- ② Evaluate $\int_1^2 x^2 dx$ using Simpson's 1/3 rule taking $h=0.2$.
3. Evaluate $\int_1^2 x^2 dx$ using trapezoidal rule taking $h=0.1$.
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[Turn Over]

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$f(x_i)$	1	2.25	3.75	4.25

8. Using Gaussian elimination method solve the following system of linear equations:

$$\begin{aligned}X + Y + Z &= 6 \\X + 2Y - 3Z &= -4 \\4X + 5Y + Z &= 17\end{aligned}$$

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Course Code: SH/CSC/503/DSE-1

Operation Research (Practical)
(OLD CBCS Syllabus 2017-2018)

Time: 2 Hrs

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Min. $Z = 2x + y$ subject to $3x + y \geq 3$, $4x + 3y \geq 6$, $x + 2y \geq 3$ and $x, y \geq 0$.

[Turn Over]

5. Solve the following LPP by revised simplex method
 $Max\ z. = x + 2y$ subject to $x + 2y \leq 3$, $x + 2y \leq 1$ and $x, y \geq 0$